

Autodesk® Scaleform®

Scaleform 4.4

/

(Scaleform 4.4 Mobile Game Kit)

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Scaleform

Nate Mitchell

1.01

2012 6 5

Autodesk® Scaleform® 4.4

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1		Error! Bookmark not defined.
1.1	Starforce Battlement	2
1.2		4
2		5
2.1		5
2.2		6
2.2.1		6
2.2.2	HUD	8
3		10
3.1	Flash	10
3.2	ActionScript	10
3.2.1		10
3.2.2	ActionScript	13
3.2.3		14

1

Scaleform

Autodesk® Scaleform®

3 , Scaleform

Starforce Battlement
(Shippable Mobile Player)

ActionScript

(PC Mac Linux iOS Android)

ActionScript 3

iOS (iOS Game Center)

Adobe® Flash®

(Asset)

Scaleform

Scaleform

Adobe Flash Player

Web

Scaleform

/

Starforce Battlement

1. Flash Professional

UI

2. ActionScript 3

(FlashDevelop 4

3. Scaleform

Exported .SWF

4. Scaleform

GPU ,

1.1 Starforce Battlement



1 Starforce – 1

/ (/)



2 Starforce –

,Starforce Battlement

/ ,

2012 5 , iOS App Store
 , Scaleform iOS

Starforce Battlement

iOS

1.2

Starforce Battlement

Scaleform

- 1.
2. GPU
3. ActionScript 3
- 4.
- 5.
- 6.
7. iOS
8. (Orientation Lock) / (Orientation Handling)

ActionScript 3

- 1.
- 2.
3. iOS
- 4.

2

2.1

- *Bin/Data/AS3/Kits/StarforceTD/* - ActionScript 3 Flash
()
- *Projects/Win32 /{Msvc80, Msvc90, or Msvc10}/Gfx 4.4 SDK /* – Visual Studio ,
Windows FxPlayer Visual Studio 2005/2008/2010 ,
Starforce.
- *Projects/iPhone/Xcode4/Gfx 4.4 iPhone SDK* – Xcode 4 , iOS
(Shippable Mobile Player) , Starforce.
- *LocalApps\StarforceTD\Android* – make Scaleform,
Eclipse , Shippable Mobile Player (Android
StarforceTD

Bin/Data/AS3/Kits/ StarforceTD Windows StarforceTD.exe
Windows Scaleform SDK

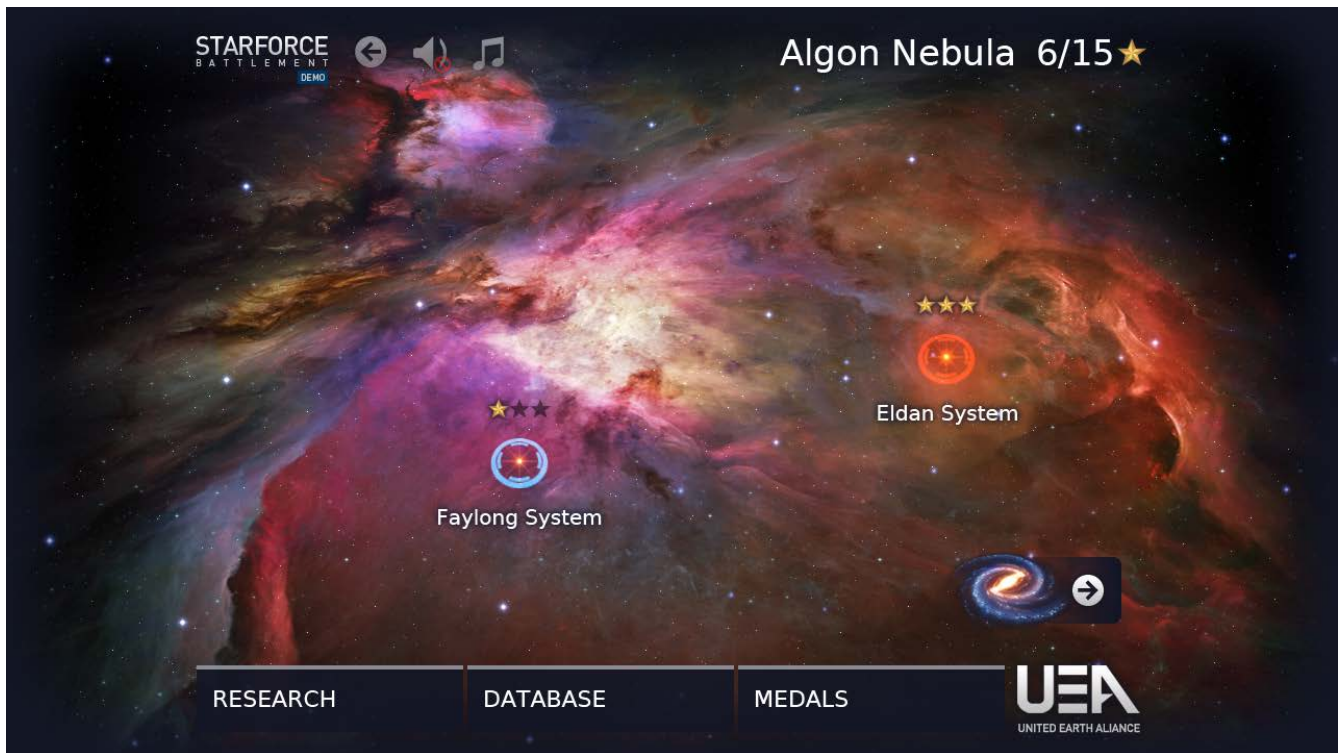
FxPlayer / FxMobilePlayer

Windows , Debugging() Working directory ()
Bin/Data/AS3/Kits/StarforceTD , Command line arguments ()
StarforceTD.swf

iOS , Xcode , iTunes SWF ()

2.2

2.2.1



3 Starforce –

StarforceTD.as StarforceTD fla Document Class(),
MainMainState MainMainState StarforceMenu.swf StarforceMenu.swf
, ActionScript 3

, / (galaxy/nebula)
2D ,



4 Starforce –

, (MissionDataView.as) (Mission
Data View) (Enter Mission)
, X (Mission Select
View)

2.2.2 HUD



5 Starforce -

Starforce Battlement

StarforceTD.swf

StarforceTD.swf

LevelGameState

-- Rush Wave ()--

HUD

1 2 3

HUD

(Health)

HUD

	:	(Orbital Strike)	(Deploy Mercenaries),
HUD		,	,
	,	(	),
	,	,	
		,	
HUD		/	/

3

3.1 Flash

Battlement Flash .FLA (Bitmap) , UI Starforce 3D (UI) FLA ActionScript

Starforce Battlement FLA

- 1. **StarforceTD fla** – .FLA Starforce Battlement UI (Main Menu) / ActionScript (Loading Screen),
- 2. **StarforceMenu fla** – (Mission Select) ActionScript StarforceTD.swf , ,
- 3. **LoadingView.swf** – () ActionScript

3.2 ActionScript

3.2.1

Starforce Battlement UI ActionScript 3 , iOS Game Center , Starforce

3.2.1.1

StarforceTD.as , Flash Document Class , StarforceTD.swf (Document Classes) , Adobe Flash

```

        'GameState'(
GameState,          StarforceTD          ,          GameState
1.  MenuGameState –
2.  LoadingGameState –
3.  LevelGameState –

StarforceTD          (          ),          StarforceMenu.swf
MainMenuView.as

```

```

    com.scaleform.std.menu
)
(Package)
(ActionScript 3
)
/
( , )
,
GameEvent( ) StarforceTD.as
LoadingGameState MenuGameState LevelGameState, LevelGameState
,
,
StarforceTD.swf LoadingGameState, LoadingGameState
HUD,
```

```
Level.as
      ,
      HUD
      ,
      ,
      )

Level.as
      Level
      (
      ,
      )

Level
      (
      ,
      )
```



```

        Scaleform
        MovieView
        MultitouchInputState,
        ,
        , Starforce
        TouchEvent
MouseEvent TouchEvent MouseEvent , TouchEvent
    MouseEvent , TouchEvent.TOUCH_TAP MouseEvent.MOUSE_CLICK
    , TouchEvent.PRESS MouseEvent.MOUSE_DOWN
    TouchEvent MouseEvent , Event(
)
    , target( )
    is ,
protected function onSetRallyClick( e:Event ):void {
    var point:Point;
    if (e is MouseEvent) {
        var me:MouseEvent = e as MouseEvent;
        point = new Point( me.stageX, me.stageY );
    }
    else {
        var te:TouchEvent = e as TouchEvent;
        point = new Point( te.stageX, te.stageY );
    }
}

```

3.2.2 ActionScript

```

/ , ActionScript ActionScript' '(Package)
com.scaleform.std
1. abilities - ( )
2. controls - UI
3. core - (State Flow) /
4. data - ,
5. entities - ,
6. events - UI
7. fx - ( )
8. hud- HUD
9. levels - ,Level( )
10. loading - (Loading Screen)

```

11. menu	–	,	UI	
12. system	–			ActionScript
13. utils	–	(	)	

3.2.3

3.2.3.1 /

```
( com.scaleform.std.system.SaveSystem.as )
```

```

ActionScript      SharedObject      SharedObject      ActionScript 3
,                  ActionScript      Scaleform      SharedObject
(Primitive Type)  (Array)

```

```

SharedObject
,SharedObject      sb_player_data

```

```

// Local reference to the galaxy progress. Populated from the PlayerData.
public static var galaxyProgress:Array = null;
// Constant String for the property of the SharedObject we'll modify.
protected static const GALAXY_PROGRESS_SO_PROP:String = "galaxyProgress";
// Reference to the SharedObject once we retrieve it.
protected static var _playerDataSO:SharedObject = null;

public static function load( pd:PlayerData ):void {
    _playerDataSO = SharedObject.getLocal("sb_player_data");
    galaxyProgress = _playerDataSO.data[ GALAXY_PROGRESS_SO_PROP ];
    pd.setGalaxyProgress( galaxyProgress );
}

public static function save( pd:PlayerData ):void {
    // Write PlayerData into SharedObject.
    _playerDataSO = SharedObject.getLocal("sb_player_data");
    galaxyProgress = pd.getGalaxyProgress();
    _playerDataSO.data[ GALAXY_PROGRESS_SO_PROP ] = galaxyProgress;
    _playerDataSO.flush();
}

```

```

, SharedObject SharedObject
SharedObject

```

```

SharedObject

```

http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/net/SharedObject.html

3.2.3.2

```

(com.scaleform.std.system.SoundSystem )
SoundEvent ( .mp3) SoundEvent
stage

```

```

dispatchEvent( new SoundEvent( SoundEvent.PLAY_SOUND, true, true,
"infantry_attack.mp3", 1, false ) );

```

```

, , SoundEvent
, SoundEvent

```

```

dispatchEvent( new SoundEvent( SoundEvent.MUTE_SOUNDFX, true, true );

```